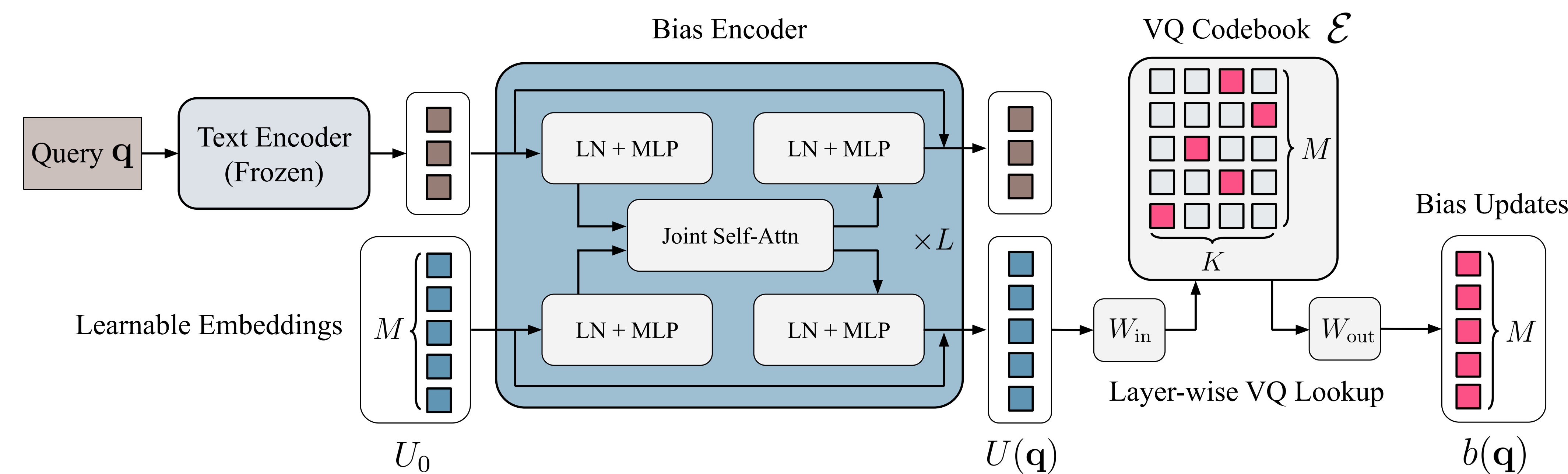


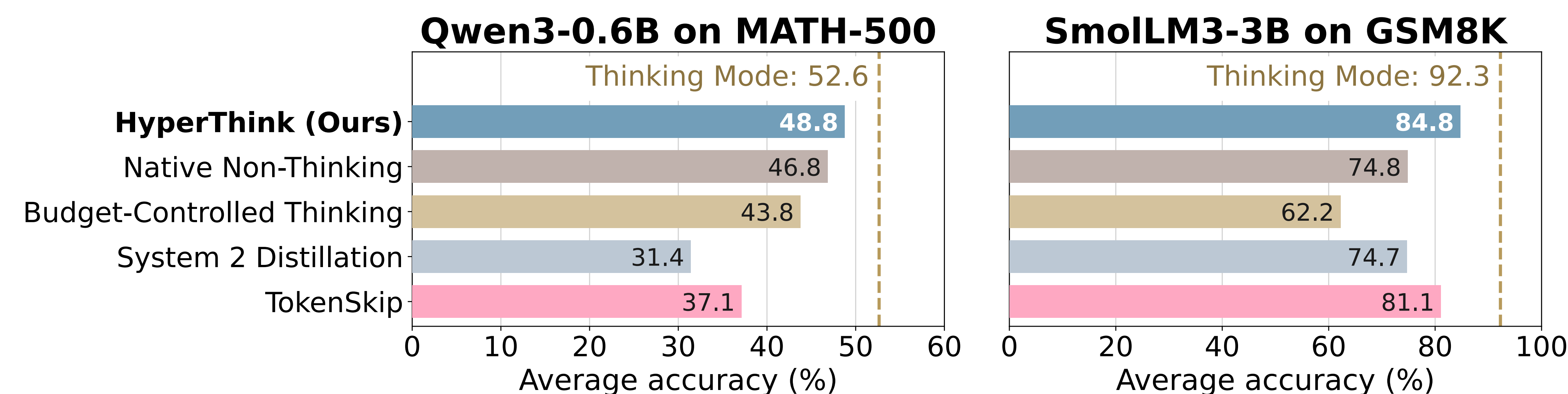
One Hypernetwork Pass Predicts a Compact Bias Update

Trained to match the LLM's answers after thinking; $\Delta\theta$ is just **0.015%** of its weights.



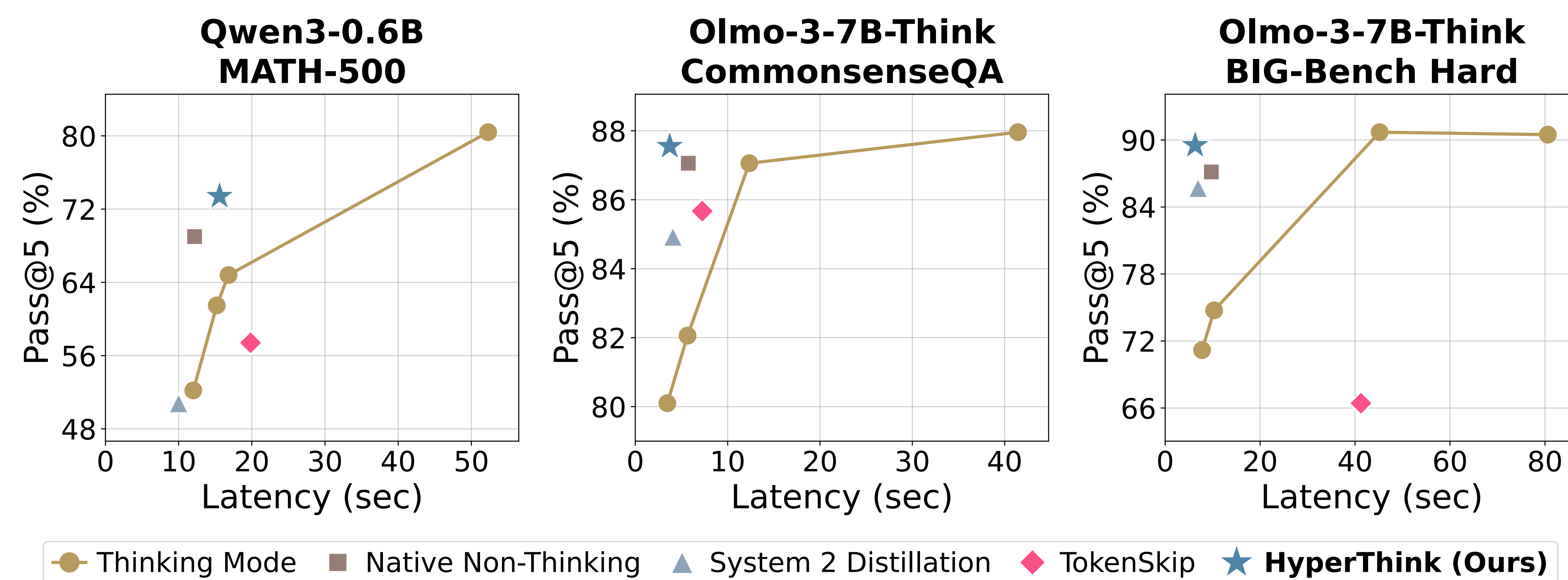
Beats Low-Budget Baselines, Even Out of Domain

The closest to Thinking Mode, at **2.5× to 3.9× fewer FLOPs**.



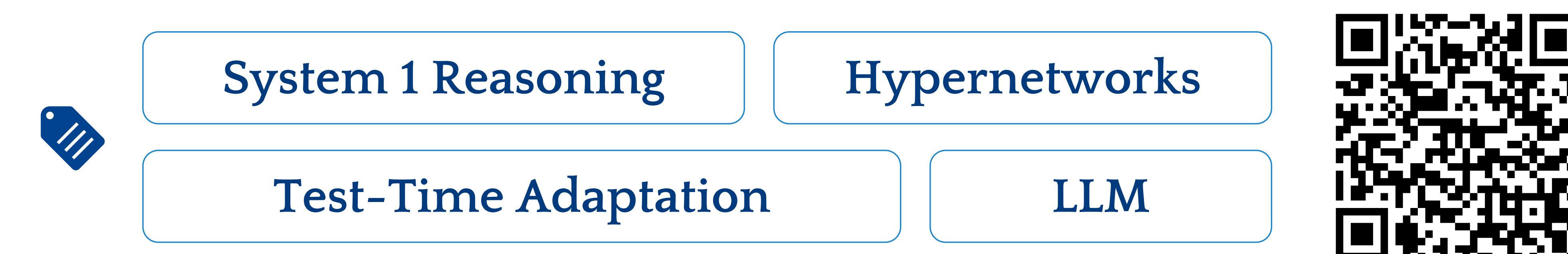
Improves the Low-Latency Trade-off

On QA, on par with full Thinking Mode Pass@5 while answering **up to 10× faster**.

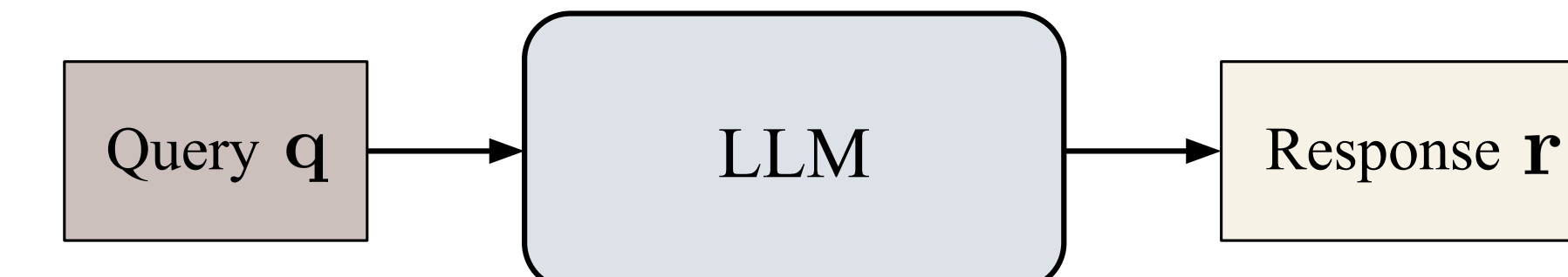


What if LLMs could think in their weights instead of their tokens?

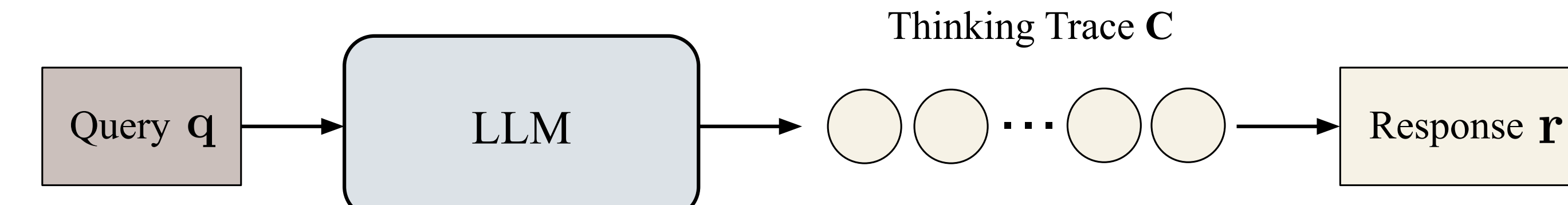
HyperThink reads the question once and predicts a **query-conditioned parameter update**; the adapted LLM then writes a short step-by-step answer with no thinking trace.



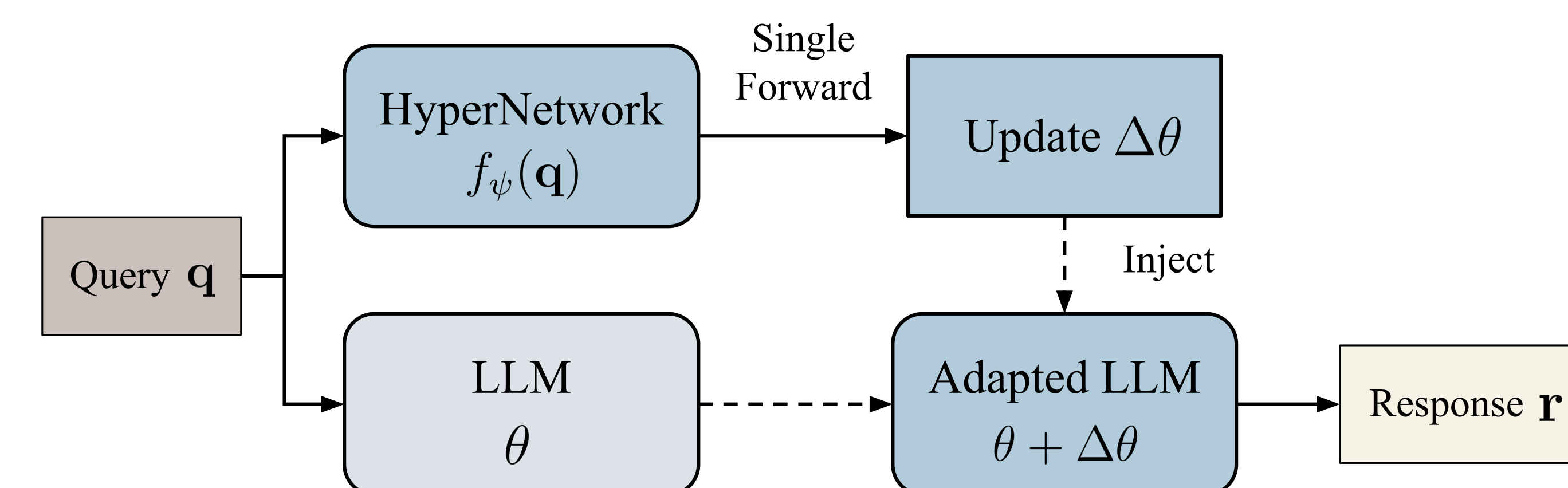
(a) Non-Thinking



(b) Thinking



(c) HYPERTHINK (Ours)

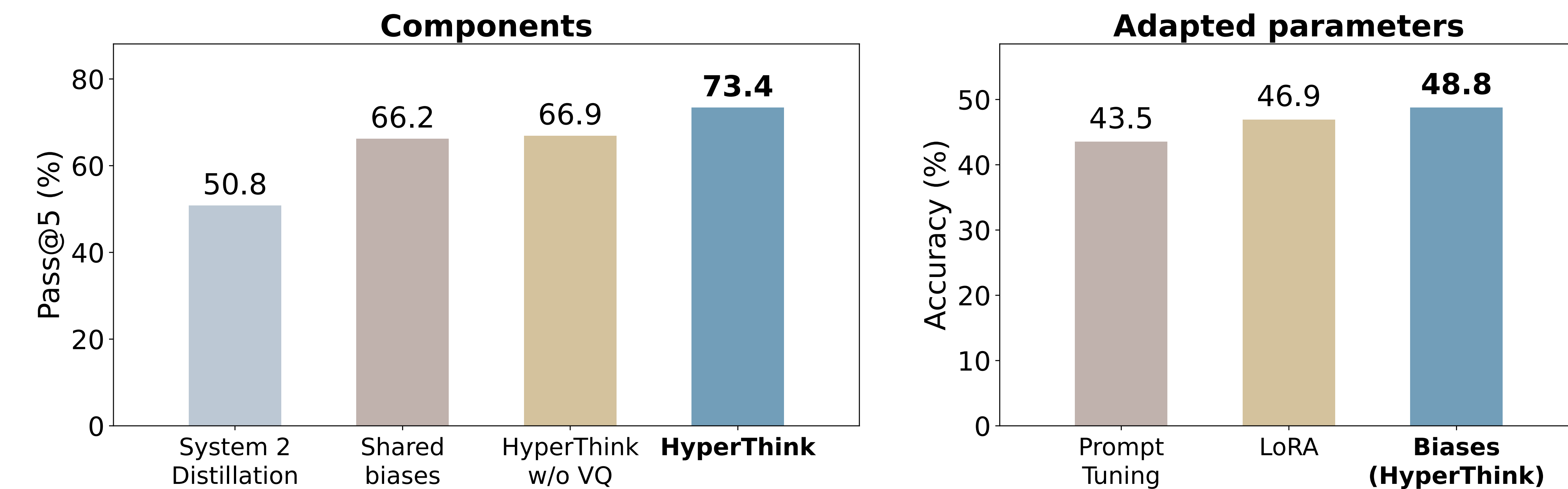


Takeaways

- Efficiency:** replaces the long thinking trace with one hypernetwork pass
- Performance:** beats Budget-Controlled Thinking on Pass@5 in every setting
- Generalization:** gains most from VQ on out-of-domain MATH-500
- Speed:** rivals full Thinking Mode on two QA tasks, up to 10× faster

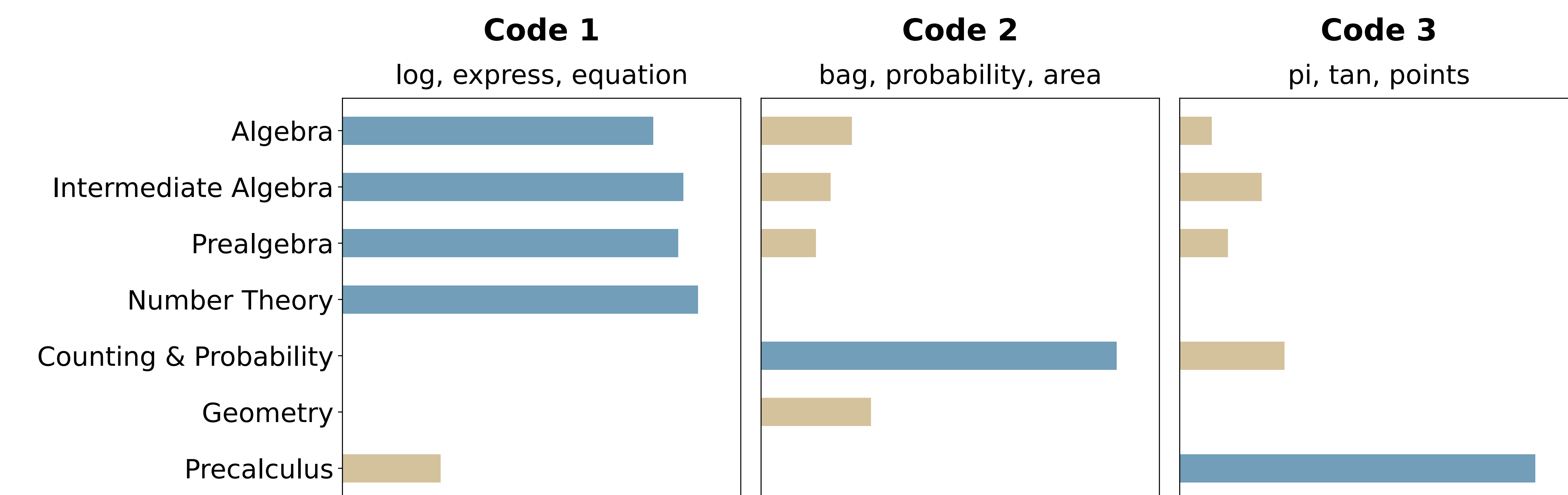
Bias Updates and VQ Both Improve MATH-500 Scores

On MATH-500, biases beat LoRA and Prompt Tuning, and **VQ adds 6.5 Pass@5**.



Each VQ Code Covers Related Math Subjects

Learned with **no subject labels**: share of each MATH-500 subject's problems per code.



Same Answer as Thinking Mode, Without the Trace

Fast like Native Non-Thinking, but **correct like Thinking Mode**.

